

Part A: Literature Review and Publicly Available Dataset

This document contains the four required tables. The first three tables summarise empirical predictive-modelling studies directly relevant to e-commerce returns. The supplied Karl systematic review was used as a background source to identify the third empirical study (Imran and Amin, 2020), because Karl is a literature review rather than a primary modelling study. The selected Part B dataset is the Synthetic E-Commerce Returns Management Dataset specified in the assignment.

Table 1 - Related literature 1

Relevance: This study directly predicts individual return events in e-commerce and demonstrates how temporal splitting, customer-product structure and strong ML baselines can be combined in a return-risk pipeline.

Title	Returnformer: A Graph Transformer-Based Model for Predicting Product Returns in E-Commerce
Harvard reference	Cao, Q., Zhang, N. and Li, H. (2026) 'Returnformer: A Graph Transformer-Based Model for Predicting Product Returns in E-Commerce', Entropy, 28(1), 72. https://doi.org/10.3390/e28010072 .
Link	https://www.mdpi.com/1099-4300/28/1/72
Modelling	Primary model: Returnformer, combining a Graph Transformer encoder, Graph External Attention (graph-level attention) and a Kolmogorov-Arnold Network (KAN) decoder for edge-level binary return prediction. Global structural information is augmented using Node2Vec embeddings and fused with original customer/product features. Large interaction graphs are partitioned into balanced-edge subgraphs. Training uses cross-entropy loss, Adam optimisation and Optuna hyperparameter search, with a 0.5 return-probability threshold and dropout. Comparison models: Multi-Layer Perceptron (MLP), XGBoost, LightGBM, CatBoost, Graph Convolutional Network (GCN), Graph Attention Network (GAT) and GraphSAGE. Evaluation used accuracy, precision, recall, F1-score and ROC-AUC.
Preprocessing techniques	The study checked for duplicates, missing values and outliers; removed duplicates; recoded customer and product IDs; converted customer birth year to age; excluded customers over 90 years old; used one-hot encoding for categorical variables; and standardised numerical variables. The temporal split was designed to reduce leakage: September-October records were used for training and October-November records for testing. The final sets contained 939,537 training events and 858,526 test events.
Datasets used	A real-world ASOS clothing e-commerce dataset covering customer keep/return events from September-November 2021. The data were organised into event, customer and product tables, with about 1.8 million keep/return events before final preprocessing, more than 1.08 million users and more than 300,000 product variants. Returned events were coded 1 and kept events 0. Sensitive information was anonymised. The paper does not provide a public download link for the ASOS data.
Key takeaways	1. Prevent leakage with time-aware splitting: because the model is intended for future return prediction, future-period information should not enter training features. 2. Use relational/interaction information when available: repeated customer-product behaviour can add predictive signal beyond isolated order attributes. 3. Use strong tabular baselines before complex graph models: for a 5,000-row synthetic tabular dataset, tree-based models such as XGBoost/LightGBM/CatBoost provide practical baselines; graph methods would only be justified if user-product network structure is explicitly modelled.

Table 2 - Related literature 2

Relevance: This study is particularly useful for preprocessing, class-imbalance handling, leakage prevention and interpretable ensemble modelling. It uses a public Kaggle dataset, satisfying the assignment requirement for a publication based on publicly accessible data.

Title	Predictive Modeling and Reverse Logistics in Fashion Product Returns - A Machine Learning and Explainability Approach for Sustainable Fashion E-Commerce
Harvard reference	Varma, M. (2025) Predictive Modeling and Reverse Logistics in Fashion Product Returns - A Machine Learning and Explainability Approach for Sustainable Fashion E-Commerce. CBDS 699 - Applied Behavioral Research Project, Harrisburg University of Science and Technology. Available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5380890 (Accessed: 28 September 2026).
Link	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5380890
Modelling	Supervised ensemble classification using Random Forest, Gradient Boosting and XGBoost. The study handled class imbalance using SMOTE applied only within training folds plus class weighting (class_weight=balanced for Random Forest/Gradient Boosting and scale_pos_weight for XGBoost). Hyperparameters were optimised using Optuna with a Tree-structured Parzen Estimator approach and more than 50 trials per seed. Five independent random seeds were used. SHAP was used for model interpretation and VADER for sentiment extraction from review text. The reported 3-way split was 60% training, 20% validation and 20% testing.
Preprocessing techniques	Missing observations were removed for key variables such as Review Text, Rating and Recommended IND. Rare Class Name levels below 1% were combined into an "other" category. Continuous variables (Age, Positive Feedback Count and Review Length) were examined for outliers using +/-3 standard deviations; log transformation was considered for highly skewed variables and implemented inside the cross-validated pipeline rather than before the split. Categorical variables were one-hot encoded inside a scikit-learn ColumnTransformer. Review Length was engineered from character count and also binned by empirical quantiles. VADER generated positive, negative, neutral and compound sentiment scores; the compound score was used as a continuous feature. SMOTE was restricted to training folds to reduce leakage.
Datasets used	Women's Clothing E-Commerce Reviews dataset, retrieved from Kaggle and attributed in the paper to the Kaggle user Nicapotato. The source paper describes 23,486 anonymised entries containing review text, product metadata, ratings and a binary Recommended IND variable. Recommended IND (1 = recommended; 0 = not recommended) was used as a proxy for return behaviour rather than a direct return label. The paper reports class imbalance of roughly 80% recommended vs. less than 20% non-recommended and later reports 19,662 complete observations after removing missing data; another section reports 19,668 usable observations, so the paper contains a small internal count discrepancy.
Key takeaways	1. Leakage control is essential: feature engineering, encoding, resampling and tuning should be kept inside the training/CV pipeline. 2. Accuracy alone is insufficient for imbalanced return classification: include F1, precision, recall and ROC-AUC, and consider a metric that reflects the business cost of missing high-risk returns. 3. Interpretability should be built into the workflow: SHAP can connect model predictions to actionable predictors. For the selected synthetic dataset, this can be applied to numeric/categorical order features without adding text sentiment features that the dataset does not contain.

Table 3 - Related literature 3

Relevance: This study is closely aligned with order-level return classification and uses order attributes such as location and payment method, which have direct analogues in the selected Part B dataset.

Title	Predicting the Return of Orders in the E-Tail Industry Accompanying with Model Interpretation
Harvard reference	Imran, A.A. and Amin, M.N. (2020) 'Predicting the Return of Orders in the E-Tail Industry Accompanying with Model Interpretation', Procedia Computer Science, 176, pp. 1170-1179. https://doi.org/10.1016/j.procs.2020.09.113 .
Link	https://www.sciencedirect.com/science/article/pii/S1877050920320135
Modelling	The study compared a traditional Decision Tree with four state-of-the-art models: XGBoost, CatBoost, LightGBM and TabNet. Hyperparameters were tuned using distributed Bayesian optimisation. The dataset was treated as a binary classification problem. Because of class imbalance, a balanced version was created using random under-sampling, producing 3,070 instances with 1,535 per class. The paper also used t-SNE as an unsupervised visual-analytics step. Evaluation included TNR, FPR, FNR, TPR/recall, ROC-AUC, F2-score, precision and accuracy; the authors emphasised TPR, ROC-AUC and F2-score because the business objective was to avoid missing high-return orders.
Preprocessing techniques	The raw dataset contained 12 columns including the target. Seven time-derived features were created from order and confirmation timestamps: order hour, order minute, order day name, confirmation hour, confirmation minute, confirmation day name and order-confirmation time gap. Redundant fields such as order_id, order_date and confirmation_date were then dropped. The paper states that the remaining seven numeric features were power-transformed to make them more Gaussian-like. Random under-sampling was applied to create a balanced dataset. The accessible paper text does not clearly specify a separate categorical encoding procedure, so no encoding method is inferred here.
Datasets used	A real-world order-return dataset from a general e-commerce retailer in Bangladesh. The raw data had 12 columns including the return target and included fields such as order date, confirmation date, order type, order location, order medium, cart order, promotional order, order amount and payment method. The paper does not identify a public download source for this retailer dataset. In the return analysis, 1,535 returned examples were reported and the balanced resampled dataset contained 3,070 observations.
Key takeaways	1. Order-level attributes are valuable: the paper highlights order location and payment method as influential, which directly motivates using User_Location, Payment_Method and other order attributes in Part B. 2. Feature engineering can add signal: when a timestamp exists, derive meaningful temporal features (for example month, day-of-week or time period) while avoiding leakage from information that would only be known after the outcome. 3. Match metrics to the business cost: use recall/F1/ROC-AUC and consider F2 where missing a true return risk is more costly than a false alarm.

Table 4 - Publicly available dataset for Part B

Relevance to Part B: The dataset contains an explicit binary return outcome and a mixture of numeric and categorical order/customer attributes, making it suitable for supervised classification after leakage-aware feature selection.

Name	Synthetic E-Commerce Returns Management Dataset
Harvard reference	sowmihari (n.d.) Synthetic E-Commerce Returns Management Dataset. Kaggle. Available at: https://www.kaggle.com/datasets/sowmihari/returns-management (Accessed: 28 September 2026).
Link	https://www.kaggle.com/datasets/sowmihari/returns-management/data
Number of instances	5,000 order-level records in a single dataset file. No predefined train/validation/test split is identified in the accessible dataset descriptions, so Part B should create its own split(s) during modelling.
Number of feature columns	22 original predictor columns plus 1 target column = 23 columns in total. The 22 predictor columns include identifiers, date, product, customer, payment/shipping, return-related, cost and sustainability fields. Not all 22 should necessarily be used as modelling features because some are identifiers or post-outcome variables.
Number of classes (if applicable)	2 classes for Return_Status: Returned and Not Returned. This is therefore a binary-classification dataset when Return_Status is used as the target.
Target range (if applicable)	Not applicable. This is a classification problem rather than a regression problem.
Origin	The Kaggle dataset is explicitly identified as synthetic and represents simulated e-commerce returns-management records rather than a disclosed real retailer transaction system. Public descriptions of the same Kaggle file report 5,000 rows and 23 columns, with order dates spanning 2022-2025 and variables covering orders, products, customer attributes, return status/reasons, return costs, profit/loss and sustainability indicators (CO2 emissions, packaging waste, CO2 saved and waste avoided). A public project using the same Kaggle dataset reports no missing values and flags several fields as post-outcome/leakage variables. Important limitation: the exact random-data generation algorithm (for example, the distributions or rules used to simulate each column) is not documented in the accessible Kaggle listing, so a more specific generation method should not be claimed without the dataset author's documentation.
Part B modelling note	For a return-risk classifier, Return_Status is the natural target. Potential predictors available before the return outcome include order date, product category, product price, quantity, discount, payment method, shipping method, user age, gender and location. Fields such as Return_Reason, Days_to_Return, Return_Cost, Profit_Loss, CO2_Saved and Waste_Avoided are generated after or because of the return outcome and should be excluded from a pre-outcome predictive model to avoid target leakage. Public documentation using the same dataset also excludes these fields for this reason.